

Training Day – 9

Today's training focused on **AWS Identity and Access Management (IAM)** and its role in securely controlling access to AWS resources. The session provided an understanding of users, groups, roles, policies, permissions, and security practices used to protect cloud environments.

Work Undertaken:

- Studied the fundamentals of **AWS IAM** and understood how it helps manage secure access to AWS resources.
- Practiced **creating and managing IAM users** and learned how individual users can be provided with controlled access to cloud resources.
- Explored **IAM groups** and understood how users can be organized into groups to simplify permission management.
- Learned about **IAM roles** and their purpose in providing permissions to AWS services and other trusted entities without directly sharing user credentials.
- Studied **IAM policies** and understood how they define which AWS resources users or other entities can access and which actions they are allowed to perform.
- Learned about the **Principle of Least Privilege**, which involves granting only the permissions required to perform a specific task, thereby reducing security risks.
- Explored **Multi-Factor Authentication (MFA)** and understood how it provides an additional layer of security beyond passwords.
- Reviewed other **AWS security best practices** for protecting cloud resources and managing access securely.

Additional Learning:

The session highlighted the importance of properly managing identities and permissions in cloud environments. Understanding IAM users, groups, roles, and policies provided a clearer picture of how organizations control access while maintaining security and minimizing unnecessary permissions.

Conclusion:

The day provided a strong foundation in **AWS security and access management**. Learning IAM concepts, the principle of least privilege, MFA, and security best practices improved my understanding of how cloud resources can be protected against unauthorized access.